

Civics Group	Index Number	Name (use BLOCK LETTERS)
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**ST ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIMS**

H1 BIOLOGY

8876/2

Paper 2: Structured & Essay Questions

Monday

10 Sept 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided in the question booklet.

Section B (Essay Question)

Answer **one** essay question.

Write your answers in the spaces provided in the question booklet.

All working for numerical answers must be shown.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/13
2	/12
3	/9
4	/11
Sub-total	/45
5/6	/15
Total	/60

Conceptual error (C)	Expression problem (E)	Misreading the question (Q)	Data quoting problem (D)

This document consists of **17** printed pages.

[Turn over

Section A

Answer **all** questions.

QUESTION 1

Fig. 1.1 shows Process X in an eukaryotic cell which produces ribosomal RNA (rRNA).

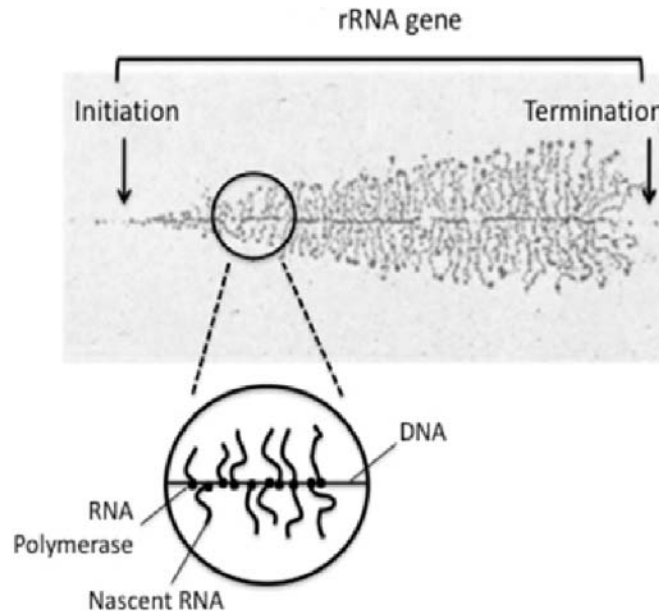


Fig. 1.1

(a)(i) Name the Process X occurring in **Fig. 1.1**.

.....[1]

(ii) List one molecule **not mentioned in Fig. 1.1** that is required for Process X.

.....[1]

(iii) Suggest how RNA polymerase is able to recognise and bind to the promoter on DNA and not to other DNA regions.

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.....[2]

(iv) Explain for the observed pattern of Process X in **Fig. 1.1**.

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.....[2]

(v) State the roles of rRNA in protein synthesis.

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.....[2]

(b) During protein synthesis in cells of an embryo, all tRNA molecules with UAC anticodon sequence, are observed to be bound to the arginine amino acid instead of methionine.

(i) Suggest how these tRNA molecules attached with the wrong amino acid might arise.

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.....[2]

(ii) Suggest and explain the effect of this wrong pairing of amino acid to tRNA on the embryo.

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.....[3]

[Total: 13]

QUESTION 2

(a) About one third of the injuries to racehorses involve tendon damage.

In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells (usually around 10 million cells for one tendon) were then injected into its damaged tendons.

80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

(i) Explain how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing the tendon injuries.

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.....[2]

(ii) Suggest an advantage of the above stem cell therapy.

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.....[1]

(iii) Similar to stem cells, cancer cells are also capable to dividing many times via mitosis. List 2 differences between stem cells and cancer cells.

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.....[2]

(b) In a dihybrid inheritance, gene B/b codes for flower colour while gene H/h codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then undergo a test cross.

(i) Predict the expected phenotypic ratio in the F2 progeny.

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.....[1]

(ii) Identify and explain which are the dominant traits in this dihybrid cross.

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.....[2]

(iii) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross if inheritance is Mendelian. [3]

(iv) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

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.....[1]

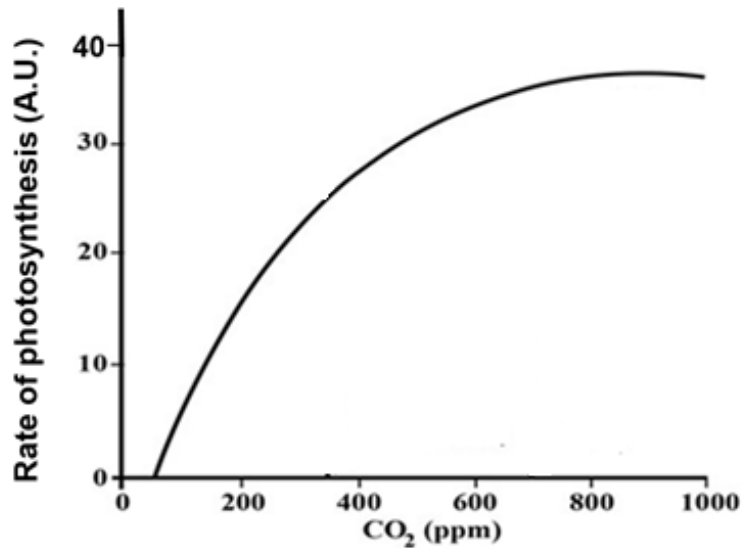
[Total: 12]

QUESTION 3

(a) In response to climate change, the rising carbon dioxide level has affected almost all crucial biological processes, including photosynthesis, respiration, and antioxidant systems, as well as other key secondary metabolisms in plants.

An experiment was designed to measure the rate of photosynthesis at a range of carbon dioxide concentrations provided to the plant.

The following results were obtained:



(i) Explain the experimental results obtained in relation to CO₂ being a limiting factor of photosynthesis.

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..... [2]

(ii) Suggest why the rate of photosynthesis reaches a plateau after a while, given that sufficient light was provided.

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..... [1]

(iii) NADP^+ is an important electron carrier found in plants, but its level decreases sharply in the day. Suggest the significance of the decrease of NADP^+ during the day.

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(b) Central to the energy metabolism of aerobically respiring cells is the enzyme ATP synthase. It consists of multiple subunits, coded for by several genes.

The enzyme is located on the inner mitochondrial membrane and catalyses the reversible reaction of ATP synthesis from ADP in intact mitochondria. Here, the **proton motive force** is required to drive the ATP synthesis.

Explain how the structure, including protein components, of the inner mitochondrial membrane is significant in driving the reaction of ATP synthase towards ATP synthesis during aerobic respiration.

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..... [4]

[Total: 9]

QUESTION 4

(a) Climate change, in the form of global warming, is expected to have an impact on various organisms. This impact however, varies geographically, particularly for insects.

Fig. 4.1 and **Fig. 4.2** below show the fitness curves of representative insects from temperate and tropical locations respectively.

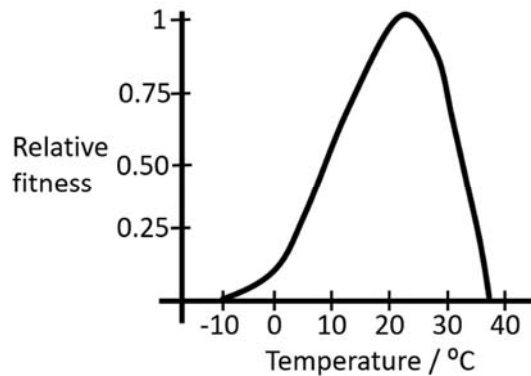


Fig. 4.1: Fitness curve of representative insect from temperate location

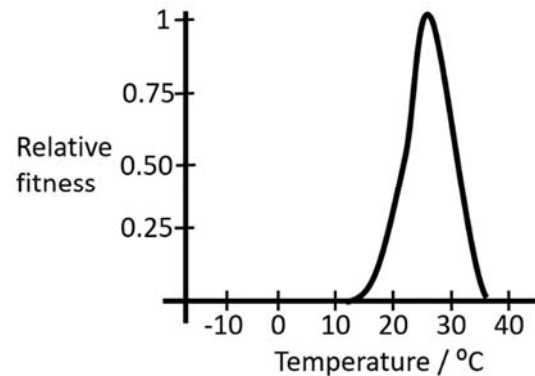


Fig. 4.2: Fitness curve of representative insect from tropical location

The mean annual temperature of the temperate location is 11°C, with a typical temperature range from 1°C to 20°C.

On the other hand, the mean annual temperature of the tropical location is 27°C, with a typical temperature range from 21°C to 31°C.

(i) Suggest what is meant by the term relative fitness in **Fig. 4.1** and **Fig. 4.2**.

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.....[2]

- (ii) With reference to **Fig. 4.1** and **Fig. 4.2** and the information given, predict and explain which insects, from the temperate or tropical location, would face a greater extinction risk as a result of global warming, assuming the warming elevates temperatures equally at both locations.

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.....[4]

- (iii) Suggest one strategy that the insects in (ii) can employ to reduce the impact of global warming on themselves.

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.....[1]

(b) Carbon dioxide is one of the greenhouse gases that contribute to global warming and subsequently climate change.

A group of high school students decided to test whether varying temperatures would correspondingly affect the mean carbon dioxide gas emission.

Table 4.1 shows the mean carbon dioxide gas emission after exposing a fixed number of mealworms to different temperatures. Carbon dioxide gas emission was measured before and after exposure to experimental temperature.

Table 4.1 showing effects of varying temperatures on mean carbon dioxide gas emission, measured in parts per million (ppm).

Temperature / °C	Mean Carbon dioxide gas emission / ppm	
	Before exposure to experimental temperature	After exposure to experimental temperature
30.0	445 ± 25	450 ± 17
40.0	450 ± 20	500 ± 30
50.0	460 ± 17	540 ± 18

Values represent mean ± standard deviation.

(i) Describe the patterns shown by the data in **Table 4.1**.

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.....[2]

(c) Corals are affected by rising temperatures in ocean waters. Explain how.

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.....[2]

[Total: 11]

Section B

Answer **one** question only.

Write your answers on the spaces provided in this question booklet.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

QUESTION 5

Cellulose is the most abundant biopolymer on earth. It forms a significant proportion of the dry mass of plants.

Outline the role of atmospheric CO₂ in contributing to the structure of cellulose in plants.
[15]

QUESTION 6

Without genetic variation, some of the basic mechanisms bringing about evolutionary change cannot operate.

Explain how mutation events may contribute to variation, and how this variation can influence natural selection.
[15]

SAJC / 8876 / H1 Biology / Prelims (2018)

(a) Stem cells undergo cell division to produce genetically identical daughter cells. Radioactive thymine was supplied to some stem cells. Fig. 1.1 shows how the amount of DNA per cell varies during periods A to E in the cell cycle.

Which monomers and types of bond are found in both glycogen and amylopectin?

- A α -glucose, glycosidic, 1,6
- B α -glucose, hydrogen, 1,4
- C β -glucose, glycosidic, 1,4
- D β -glucose, hydrogen, 1,6

The huia, *Heteralocha acutirostris*, was found in New Zealand until 1907, when it became extinct. This bird had a ground-feeding habit and was particularly noted for large, attractive tail feathers. Males and females had very different beak forms, with the males having a short strong beak, whilst the females had a long curved beak to reach into otherwise inaccessible places. What is the most likely reason for the extinction of the huia?

- A Huia fed on species introduced by humans. When these declined, the huia population fell.
- B In the face of a declining population the huia evolved into a tree-living species.
- C Male and female huia were unable to breed successfully owing to strong sexual dimorphism.
- D New competitors in New Zealand occupied part of the huia's niche.

[Total: 10]

Time

QUESTION 2 DNA

Fig 3.1 shows part of a DNA molecule.

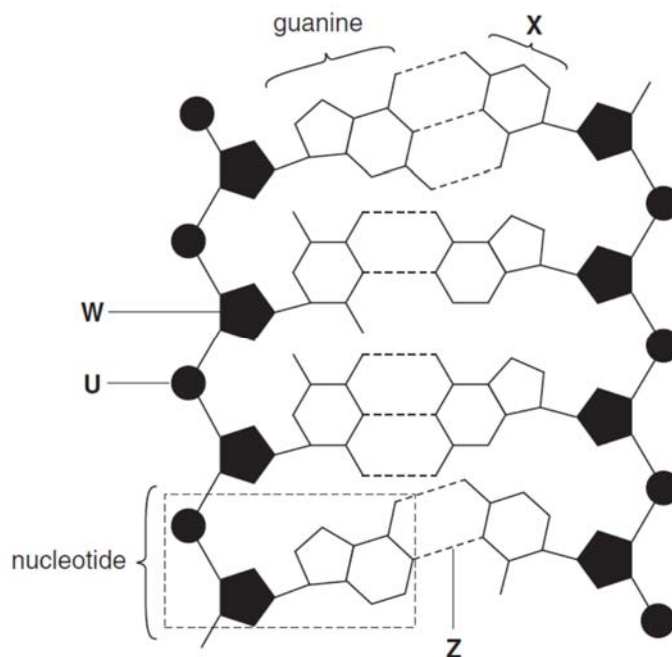


Fig 3.1

(a) (i) Name **U** to **X**.

U

W

X [1]

- 1 **U** - phosphate / PO_4 ;
REJECT phosphoric acid / phosphorus / P
- 2 **W** - deoxyribose;
REJECT pentose
- 3 **X** - cytosine;
REJECT nitrogenous base / pyrimidine / C

(ii) Explain the significance of the bonds indicated by **Z** during the DNA replication process.

.....[2]

- 1 Hydrogen bonds
- 2 Allows for **complementary base pairing** ($\text{A} = \text{T}$, $\text{G} = \text{C}$) and DNA repair / **proof-reading**/replacement of incorrectly-inserted nucleotide;

- 3 Ref. genetic stability (DNA sequence remains intact) / maintains integrity of DNA base sequence;

(iii) Describe three features of a polypeptide molecule that are different from those found in a DNA molecule.

.....[3]

	polypeptide	DNA
1	amino acids as subunits/monomers;	Nucleotides as monomers;
2	one / single strand/chain	Exists as double helix; / two/double strandd/chains
3	peptide bonds between amino acids	Phosphodiester bonds between nucleotides;
4	Has no phosphate / PO ₄	Has phosphate / PO ₄
5	Consists of secondary and tertiary structure	Helical in structure

(b) Explain

☆☆.....[3]

[Total: 12]